THE INTERNAL HOM ALONG A FINITE RING MAP CARRIES A DUALIZING COMPLEX TO A DUALIZING COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The representation of the internal hom along a finite map by a bounded complex of injectives. Invoked at the start of the counted proof (source lines 3032-3036). Gives finite injective dimension of the resulting complex.
- The preservation of bounded below coherent complexes by the internal hom along a finite map. Invoked inside the counted proof (source lines 3038-3039). Gives that the resulting complex has finite cohomology modules.
- The adjunction between the internal hom along a finite map and restriction, together with the dualizing property of the original complex. Invoked inside the counted proof (source lines 3040-3052). Computes the hom modules of the resulting complex with itself, which are the last property of a dualizing complex.

Source reference. Original lemma lemma-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 2818–2818. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-exact-support-coherent: sources/stacks-dualizing/pinned-original-source.tex, source lines 2480–2480. Navigation pointer only: the counted text refers to the Stacks item lemma-exact-support-coherent, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-hom-injective: sources/stacks-dualizing/pinned-original-source.tex, source lines 433–433. Navigation pointer only: the counted text refers to the Stacks item lemma-hom-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-right-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 2428–2428. Navigation pointer only: the counted text refers to the Stacks item lemma-right-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $A \to B$ be a finite ring map of Noetherian rings. Let $\omega_A^\bullet$ be a dualizing complex. Then $R\mathrm{Hom}(B, \omega_A^\bullet)$ is a dualizing complex for B.*

Proof. Let $\omega_A^\bullet \to I^\bullet$ be a quasi-isomorphism with $I^\bullet$ a bounded complex of injectives. Then $\mathrm{Hom}_A(B, I^\bullet)$ is a bounded complex of injective B-modules (Lemma lemma-hom-injective) representing $R\mathrm{Hom}(B, \omega_A^\bullet)$. Thus $R\mathrm{Hom}(B, \omega_A^\bullet)$ has finite injective dimension. By Lemma lemma-exact-support-coherent it is an object of $D_{Coh}(B)$. Finally, we compute

$$\mathrm{Hom}_{D(B)}(R\mathrm{Hom}(B, \omega_A^\bullet), R\mathrm{Hom}(B, \omega_A^\bullet)) = \mathrm{Hom}_{D(A)}(R\mathrm{Hom}(B, \omega_A^\bullet), \omega_A^\bullet) = B$$

and for $n \neq 0$ we compute

$$\mathrm{Hom}_{D(B)}(R\mathrm{Hom}(B, \omega_A^\bullet), R\mathrm{Hom}(B, \omega_A^\bullet)[n]) = \mathrm{Hom}_{D(A)}(R\mathrm{Hom}(B, \omega_A^\bullet), \omega_A^\bullet[n]) = 0$$

which proves the last property of a dualizing complex. In the displayed equations, the first equality holds by Lemma lemma-right-adjoint and the second equality holds by Lemma lemma-dualizing. □